A common-disorder algebra for two bilocal channels

Mixed-q moments, actual-operator transfer and a real-time four-point function

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Abstract

The charged curvature and neutral slope of the reconstructed singlet bilocal have a joint noncommutative limit. The charged field's two-species mixed-q law was already obtained in an earlier project stage. We identify its exact finite-size interface and add the neutral Clifford-size residual, using source-dependent singleton normalization and the complete local collision polynomial. The resulting three species have effective chord lengths $(1, 1, 2)$. Uniform fixed moments and strong replacement transfer this law to the actual low memory coefficients. Their alternating four-point function and product Gram matrix give a strictly positive, operator-valued commutator. An existing unequal-probe kernel then supplies its real-time evolution. The full finite-relative-time bilocal four-point function requires an additional summable fourth-norm tail; the present result specifies that next step.

1 Model, state and the two fields

Use standard CAR and the overlapping complex Hermitian Gaussian p-body Hamiltonian H_N, with the same original disorder in every charge sector. The state is normalized annealed full-Fock trace $\tau_N = \mathbb{E}_J 2^{-N} \operatorname{Tr}$ at $\beta = \mu = 0$. Take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed $J_s > 0$, $\hbar = 1$, and string time. Put

$$C = J_s^2/\lambda, \qquad q = e^{-4\lambda}, \qquad \|X\|_r = (\tau_N |X|^r)^{1/r}. \tag{1}$$

Let Q be charge, $DX = QX - \sum_i c_i^\dagger X c_i$ on neutral operators, and $\mathcal{L}\Gamma_A = |A|\Gamma_A/2$ the half-Clifford size. With $C_Q = \mathbb{E}_J H_N^2$ and $Z_2 = \mathcal{W}_2(H_N^2) = H_N^2 - C_Q$, define

$$W_{1,N} = (p - \mathcal{L} - \lambda_N) H_N, \tag{2}$$

$$J_{2,N} = (2p - D) Z_2 - \kappa_2 (H_N^2 - C), \qquad \kappa_2 = 2\lambda \frac{1-q}{1+q}. \tag{3}$$

W_1 is a field, whereas $\mathcal{W}_2$ is coupling-Gaussian chaos projection. Neither is spectral q-Wick ordering. Replacing λ_N by λ in W_1 changes every fixed norm by $o(1)$.

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2 What was already known, and its exact interface

The previous singlet-multimark stage [4] established the joint law of H and the centered charged singlet. Its fourth moment and its effective length two are reused here. The interface must hold for the same disorder, not just for two matching marginal spectra. Set

$$\mathcal{A} = \sum_i [H, c_i]^\dagger [H, c_i], \qquad O = \mathcal{A} - \mathbb{E}_J \mathcal{A}, \qquad Y^{\mathrm{old}} = O - \kappa_2 Z_2. \tag{4}$$

Since $DH = pH$, direct CAR expansion gives the finite-N identities

$$\mathcal{A} = (2p - D)H^2, \qquad O = (2p - D)Z_2, \tag{5}$$

$$J_{2,N} - Y_N^{\mathrm{old}} = \kappa_2 (C - C_Q). \tag{6}$$

$C_Q|_m = C_m I$ is a bounded deterministic charge function. Central charge asymptotics and binomial concentration give $C_Q \to C$ in every fixed L^r. Thus (6) is a strong common-sample bridge in every fixed norm.

For the actual charge-removal compression in sector $m > 0$, write $h = H_m$, $K = I_N \otimes H_{m-1}$ and

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \quad R = (1 - VV^\dagger)KV, \quad \alpha = 1 - p/m. \tag{7}$$

The block square gives

$$mR^\dagger R = \mathcal{A}|_m - \frac{p^2}{m} h^2. \tag{8}$$

Let $\mathcal{P}_H$ denote samplewise energy pinching, retaining each whole degenerate eigenspace. Charge scalars and h^2 commute with h, so the actual zero-time dynamic memory is exactly

$$\boxed{A_{0,N} = (1 - \mathcal{P}_H) m R^\dagger R = (1 - \mathcal{P}_H) J_{2,N}.} \tag{9}$$

Inactive sectors have zero dynamic field; $m = 0$ is set to zero without using the expression containing $1/m$.

The previously proved absence of diagonal spectral mass gives $\|\mathcal{P}_H J_{2,N}\|_2 \to 0$. Old fixed high moments and pinching contraction upgrade this to, for example,

$$\|A_{0,N} - J_{2,N}\|_4 \le \|\mathcal{P}_H J_{2,N}\|_2^{1/3} \|J_{2,N}\|_8^{2/3} \longrightarrow 0. \tag{10}$$

Using any higher even norm gives the same conclusion for every fixed finite r. This also recovers the old variance and fourth moment:

$$w = \|J_2\|_2^2 = C^2 \left(4\lambda + \frac{16\lambda^2 q}{1+q}\right), \qquad \tau J_2^4 = (2 + q^4) w^2. \tag{11}$$

The new question is the joint law with W_1, not a new derivation of this already recorded charged fourth moment.

3 Finite sources before taking the limit

Consider a fixed trace word of pure coupling-Wick blocks, each with one D or $\mathcal{L}$ mark on a disjoint region, and unmarked Hamiltonian insertions. Internal Gaussian pairings in a pure Wick block are forbidden. For r marks use the nilpotent algebra

$$\mathfrak{A}_r = \mathbb{R}[z_1, \dots, z_r]/(z_1^2, \dots, z_r^2). \tag{12}$$

All source masks are retained. For a single-site matrix,

$$D_i \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & b/2 \\ c/2 & d-a \end{pmatrix}, \qquad \mathcal{L}_i \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{2} \begin{pmatrix} a-d & b \\ c & d-a \end{pmatrix}. \tag{13}$$

Multiply the complete local word in each marked region before applying $1 - z_a D_i$ or $1 - z_a \mathcal{L}_i$, and preserve the CAR reordering sign. Denote its normalized trace by $T_S(z)$ for a site shared by labels S. A singleton trace $t_u(z)$ has nonzero constant term $1/2$. Exact normalization cancels the extensive marks:

$$e_{\mathrm{nil}}^{\sum_a p k_a z_a} \prod_u (2t_u(z))^p = 1, \qquad R_S(z) = \epsilon_S T_S(z) \Big/ \prod_{u \in S} t_u(z). \tag{14}$$

This division is by the source-dependent singleton, never by a possibly zero unmarked collision trace.

Uniform fixed moments. With 2ℓ fixed set labels, let $R_{\mathrm{tot}} = \sum_{u<v} |I_u \cap I_v|$ and $M = \binom{2\ell}{2}$. The coefficient norm of each local ratio is bounded by a fixed B. The full normalized source word is bounded by $B^{R_{\mathrm{tot}}}$. Hypergeometric factorial moments and Hölder give

$$\mathbb{E}\binom{|I_u \cap I_v|}{k} \le \frac{\lambda_N^k}{k!}, \qquad \mathbb{E}B^{R_{\mathrm{tot}}} \le \exp\{\lambda_+(B^M - 1)\}. \tag{15}$$

This proves every fixed marked moment and Schatten even-moment bound before discarding higher intersections. Weighted triple-intersection terms then vanish by Cauchy–Schwarz, since their probability is at most $\binom{2\ell}{3} p^3/N^2$. Joint factorial moments identify the remaining pair overlaps as independent Poisson variables.

For a Gaussian pairing π, the full limiting source weight is

$$C^\ell e^{\lambda \kappa_\pi(z)}, \qquad \kappa_\pi(z) = \sum_{u<v} (R_{uv}(z) - 1) - \ell, \qquad \kappa_\pi(0) = -4\,\mathrm{cr}(\pi). \tag{16}$$

These are the BNR local-CAR/Poisson ingredients [1], applied to actual common-disorder fields. The finite-N domination is for each fixed number of slots and marks; it is not a degree-uniform bound for an infinite time series.

4 The neutral source and the three-species law

A pair collision has four incidences. A marked region containing no incidence contributes no source. A region with one incidence contains an odd c or $c^\dagger$; its factor $1 - z_a/2$ cancels exactly in (14). Hence each surviving source needs at least two incidences. Disjoint marked regions allow at most two such sources. For this specified class, all source coefficients of degree three and higher in κ_π vanish. This is an extra local argument, not a consequence of the disappearance of triple set intersections. The complete fourth source moment is still generally nonzero:

$$[z_1 z_2 z_3 z_4] e^{\lambda(\kappa_\pi - \kappa_\pi(0))} = \sum_{\mathcal{P} \in \mathrm{Part}([4])} \lambda^{|\mathcal{P}|} \prod_{A \in \mathcal{P}} \kappa_A. \tag{17}$$

Only singleton and pair blocks survive in this sum.

For a one-slot W_1 source, only collision of its own creator and annihilator sets survives singleton cancellation. The local ratios are $2 + z_W$ with an unmarked opposite endpoint and $2 + z_W + z_{W'} + z_W z_{W'}$ with another one-slot size mark. With a whole-D mark at the opposite endpoint the ratio remains $2 + z_W$, independent of that D source. Thus

$$\kappa_W = 1, \quad \kappa_{WW'} = \mathbf{1}_{\text{same Gaussian chord}}, \quad \kappa_{WD} = 0. \tag{18}$$

Subtracting $\lambda_N H$ removes the single-source term in the limit. Every surviving W must therefore pair directly with another W, contributing $C\lambda$. Such chords are unmarked spectators for the old J_2 adjacent-endpoint cancellation [4]. The surviving charged pairs compress to length-two chords.

Theorem 1 (Fixed words of actual low generators)**.** *Define $h = H/\sqrt{C}$, $u = W_1/\sqrt{C\lambda}$ and $x = J_2/\sqrt{w}$. Their limiting normalized trace moments are*

$$\tau(a_1 \cdots a_n) = \sum_{\substack{\pi \in \mathcal{P}_2(n) \\ \text{equal-species pairs}}} \prod_{\{e,f\} \in \mathrm{Cr}(\pi)} Q_{s(e)s(f)}, \qquad Q = \begin{pmatrix} q & q & q^2 \\ q & q & q^2 \\ q^2 & q^2 & q^4 \end{pmatrix}, \tag{19}$$

where the species order is (h, u, x). The same law holds after replacing J_2 by A_0 and W_1 by $B_1/(8\lambda C q)$.

The actual neutral replacement is proved next. The matrix in (19) is an existing mixed-q crossing matrix; a positive tracial realization is provided by Junge–Zeng [3]. Our identification concerns particular same-disorder microscopic fields. It makes no assertion of independent finite-N probes.

5 Actual neutral slope and real-time transfer

The neutral first coefficient from the memory expansion is

$$B_{1,N} = m\epsilon_m\alpha_m(1-\mathcal{P}_H)B_{Z,m}, \quad B_{Z,m} = \Phi_m(h) - \bar{\kappa}_m h, \quad \epsilon_m = 1-\alpha_m^2. \tag{20}$$

Here Φ_m and $\bar{\kappa}_m$ retain their exact deterministic sector definitions from the two-component stage [5]. Its full-charge L^2 bridge to $8\lambda CqW_1$ is already known. To transfer four-point products one needs higher norms on both sides.

The unpinched $T_{1,m} = m\epsilon_m\alpha_m B_{Z,m}$ is a Hermitian, $U(N)$-equivariant linear Gaussian field. Irreducibility of $\bigwedge^m \mathbb{C}^N$ gives $\mathbb{E}T_{1,m}^2 = v_m I$. For any such $X = \sum g_a A_a$, Gaussian integration by parts, followed by weighted AM–GM in the eigenbasis of X, gives

$$\mathbb{E}\,\mathrm{tr}_m\, X^{2k} \le (2k-1)v_m \mathbb{E}\,\mathrm{tr}_m\, X^{2k-2} \le (2k-1)!!\, v_m^k. \tag{21}$$

The previous broad-central estimate

$$\|B_{Z,m}\|_2 \le 12C_m^{3/2}(p/N)\sqrt{K_1}, \qquad m\epsilon_m \le 2p \tag{22}$$

bounds v_m uniformly there. Polynomial coarse bounds in the remaining sectors are dominated by the binomial charge tail. Pinching is contractive and $\|1-\mathcal{P}_H\|_{r\to r} \le 2$. Thus B_1 has all fixed high moments. The local-source proof supplies them for W_1; interpolation with the established L^2 bridge yields

$$\boxed{\|B_{1,N} - 8\lambda CqW_{1,N}\|_r \longrightarrow 0 \quad (r<\infty \text{ fixed}).} \tag{23}$$

This argument uses linear Gaussian covariance, not a general matrix hypercontractivity claim for higher rank descendants.

For four factors, telescoping and tracial Hölder give

$$|\tau(X_1X_2X_3X_4 - Y_1Y_2Y_3Y_4)| \le \sum_j \|X_j - Y_j\|_4 \prod_{i\neq j} \max(\|X_i\|_4, \|Y_i\|_4). \tag{24}$$

Equations (9) and (23) therefore transfer all fixed words to the actual coefficients. For center-time evolution, finite unitary insertions preserve norms. Polynomial moment convergence extends to these insertions: bounded C_m and (21) imply $\|H_N\|_s \le K\sqrt{s}$, so the Taylor remainder satisfies

$$\left\| e^{iTH_N} - \sum_{j=0}^{k} \frac{(iTH_N)^j}{j!} \right\|_r \le \frac{|T|^{k+1}(K\sqrt{r(k+1)})^{k+1}}{(k+1)!} \to 0. \tag{25}$$

Higher Hölder exponents handle the fixed field insertions. This is uniform on compact center-time intervals and proves real-time joint moments for the two fixed generators.

6 Noncommutativity and the existing real-time kernel

Equation (19) gives

$$\tau(xu) = 0, \qquad \tau(x^2u^2) = 1, \qquad \tau(xuxu) = q^2, \tag{26}$$

$$\tau(u^4) = 2 + q, \qquad \mathrm{Gram}(xu, ux) = \begin{pmatrix} 1 & q^2 \\ q^2 & 1 \end{pmatrix}. \tag{27}$$

The Gram determinant is $1 - q^4 > 0$. In particular

$$\boxed{\|[x, u]\|_2^2 = 2(1 - q^2) > 0, \qquad \|[J_2, W_1]\|_2^2 = 2wC\lambda(1 - q^2).} \tag{28}$$

The commutator has zero trace and nonzero norm, so it cannot be a scalar operator. Two-point orthogonality has not made these fields commute. After the double-scaled limit, its normalized equal-time squared norm is $16\lambda + O(\lambda^2)$ for small λ.

For $x(T) = e^{iTH}xe^{-iTH}$ define

$$F_{xu}(T) = \tau[x(T)u\, x(T)u], \qquad D_{xu}(T) = \|[x(T), u]\|_2^2. \tag{29}$$

The uncrossed mixed channel is $\tau[x(T)^2u^2] = 1$: its two adjacent same-species pairs cross no Hamiltonian chord, leaving cancelling propagators. Thus $D_{xu} = 2(1 - F_{xu})$. Only one equal-species crossed pairing is allowed. Use the BINT series [2] with $a = q^2$, $b = q$, and restore the mutual-probe weight $Q_{xu} = q^2$. In symmetric Jacobi normalization put $\sigma = \sqrt{C}$ and

$$J_{n-1,n} = \sigma\sqrt{\frac{1 - q^n}{1 - q}}, \qquad U(T) = e^{-iTJ}, \qquad v_n = U_{n0}(T), \tag{30}$$

$$c_i = \frac{(b^2; q)_i}{(q; q)_i}a^i, \qquad V_{li} = \sqrt{c_i}\, b^l\sqrt{\frac{(q; q)_{l+i}}{(q; q)_l}}v_{l+i}. \tag{31}$$

With $(z; q)_i = \prod_{j=0}^{i-1}(1 - zq^j)$, the answer is

$$\boxed{F_{xu}(T) = q^2\sum_{n,i\geq 0} a^n|(U(T)^\dagger V)_{ni}|^2.} \tag{32}$$

The formula is inherited; identifying the actual fields is the new interface. Independent colored-chord Taylor enumeration gives

$$F_{xu}(T) = q^2 - 2q^2(1 - q)^2(1 + q)(\sigma T)^2 + c_4(q)(\sigma T)^4 + O(T^6), \tag{33}$$

$$c_4(q) = \frac{q^2(q - 1)^3(q + 1)}{6}(9q^3 + 9q^2 - 3q - 11). \tag{34}$$

In particular $F_{xu}(0) = q^2$. Removing the mutual factor would incorrectly erase the equal-time commutator (28).

7 Numerical four-point function and its errors

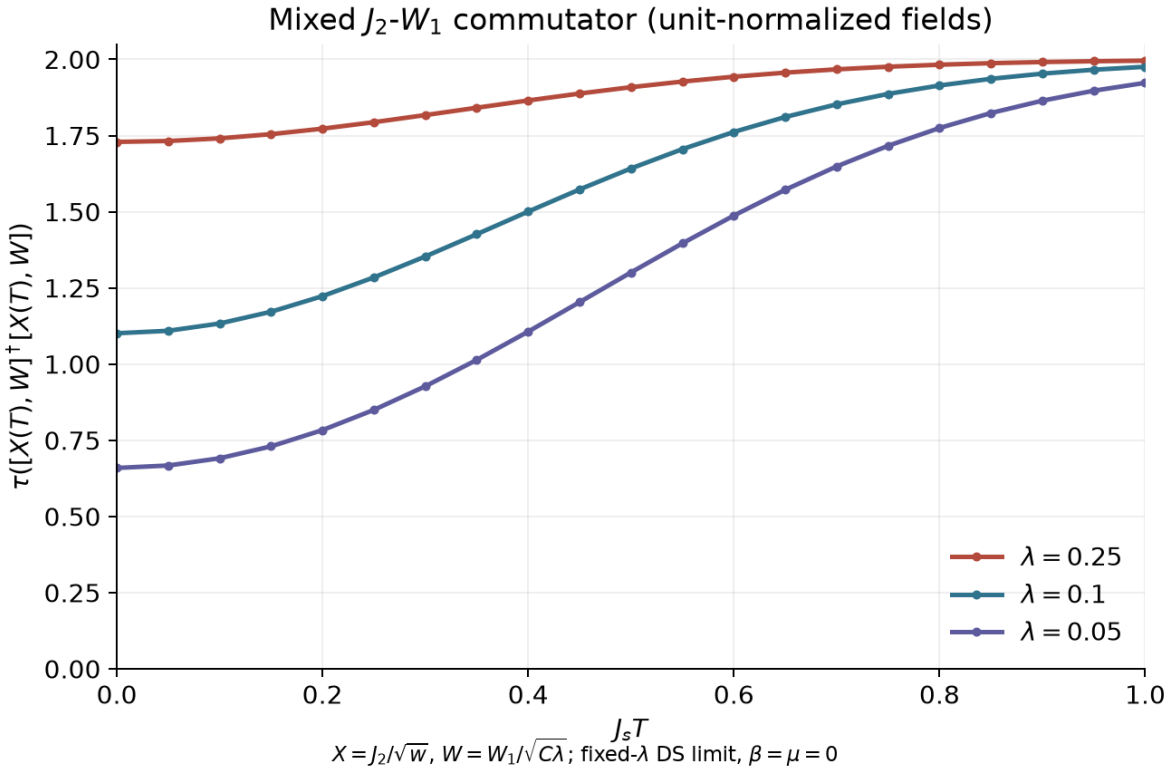


Figure 1: Real-time commutator of the two unit-normalized low channels. The double-scaled limit is taken first at each fixed λ; $J_s = 1$. The curves do not assert a full bilocal four-point limit.

λ	Jacobi cutoff L	$F_{xu}(1)$	$D_{xu}(1)$
.25	61	.00202844	1.99594313
.10	130	.01212498	1.97575004
.05	245	.03842489	1.92315023

There are 22 times per parameter, $0, .05, \dots, 1$ plus .01. Each value is compared with cutoff $L + 16$ and with exchanged probe lengths $(1, 2)$. The largest cutoff and species-swap changes are 1.89×10^{-15} and 2.23×10^{-16}. Independent exact Taylor coefficients through order six use 130 arc layouts and 1,376 colored pairings per species order. The small-time numerical comparison differs by at most 1.45×10^{-15}.

For analytic truncation, let $C_b = (1-q)^{-1}$, $K_b = C_b/(1-a)$, $R_H = 2\sigma/\sqrt{1-q}$ and $T_k(z) = \sum_{j \geq k} z^j/j!$. The source map obeys $\|v \mapsto V\| \leq \sqrt{K_b}$, with source rows beyond M bounded by $\sqrt{K_b} b^M$. Jacobi path agreement below the boundary and unitarity give

$$|F_{xu} - F_{xu}^{(L)}| \leq 2q^2 K_b \{2T_L(R_H|T|) + 2T_{L-M}(R_H|T|) + 2b^M\}. \tag{35}$$

The recorded interval truncation bounds are below 5.44×10^{-12}. They control truncation in exact arithmetic; the floating comparisons are not interval certificates. Saved data include relative errors and the subtraction condition $(1 + |F|)/|1 - F|$, whose maximum here is about 5.067.

8 A stronger return estimate and the next target

The previous reconstruction [6] writes the actual dynamic bilocal as a linear filter of its actual memory, plus a controlled return error:

$$\widetilde{\mathcal{B}}_N(t) = -\int_0^t ds \int_{-(t-s)/2}^{(t-s)/2} du\, e^{iHu}\mathcal{F}_N(s)e^{-iHu} \quad (t \geq 0). \tag{36}$$

Negative times use the adjoint relation. Its contractive background propagator F_0 obeys the exact ordered identity $F - F_0 = -F_0 * \delta M * F$. The old force moment theorem supplies every fixed high norm, not just the eighth moment used in that stage. For fixed $r \geq 2$, in the good charge region $m \geq N/4$,

$$\|\delta M\|_2 = O(p^{-2}), \quad \|\delta M\|_{4r} = O(p^{-1}), \quad \|F - U_H\|_{2r} + \|F_0 - U_H\|_{2r} = O(p^{-1}). \tag{37}$$

Interpolation with $\theta = 1/(2r-1)$ gives $\|\delta M\|_{2r} = O(p^{-1-\theta})$. Hölder in the ordered return integral, $m \leq N = O(p^2)$, and the bad-charge binomial tail therefore yield the direct extension

$$\sup_{|t|\leq S} \|\mathcal{B}_N(t) - \widetilde{\mathcal{B}}_N(t)\|_r \leq K_{r,S}\{p^{-1/(2r-1)} + pe^{-N/(8r)}\}. \tag{38}$$

At $r = 4$ this gives $O(p^{-1/7})$. It controls the return itself.

To pass the complete filtered field to four-point products, the next task is a summable fourth-norm tail, for instance

$$\lim_{K\to\infty} \limsup_N \sup_{|t|\leq S} \|\widetilde{\mathcal{B}}_N(t) - \widetilde{\mathcal{B}}_N^{(K)}(t)\|_4 = 0, \tag{39}$$

with a specified finite field/degree truncation. Fixed-degree bounds from (16) can grow with the degree; the old L^2 tail does not yet prove (39). Thus actual low-channel real-time four-point convergence is established here, while the full finite-relative-time bilocal four-point function remains open.

The physical next step is to complete that algebra and determine its state and polarization before comparing with canonical meson dressing. Itakura's construction [7] uses vacuum normal ordering, momentum polarization and a restricted physical subspace. The current trace and its non-scalar commutator are concrete data for that comparison. No microscopic P^-, internal quark momentum fraction, bulk mass assignment or finite-G gravitational interaction is identified by the present low-channel OTOC.

9 Evidence and reproducibility

Four new scientific outputs are saved without overwriting previous results. Analytical proofs extend the finite checks to fixed-word limits; finite fixtures are not double-scaling fits.

- Complete local sources: 40,032 exact finite CAR/source coefficient comparisons, with all sixteen masks for four marks; 4,768 collision exponent comparisons and a separate partition expansion. There are 1,490 verified zero cubic/quartic exponent coefficients and 216 nonzero full four-mark coefficients. Negative local contributions are retained.
- Joint moments: 53 cyclic representatives cover 233 words, including 23 zero representatives, four old J_2/H interface checks, and 1,763 accepted diagrams. Exact source polynomials are compared with the compressed colored-pairing rule. Formal rational (q, λ) checks do not impose $q = e^{-4\lambda}$ and are algebraic diagnostics.
- Actual finite interface: three full-Fock fixtures at $(N, p) = (2, 1), (3, 2), (4, 2)$ give 33 exact matrix identities and 3,030 entry comparisons using rational arithmetic. The last has nonzero commutator norm squared 73567/144; the first two are zero controls. Deterministic real Hermitian couplings, rather than a Gaussian sample or a limiting normalization, test these algebraic identities.
- Real-time kernel: 66 time points, 198 kernel evaluations, two cutoff sizes and species exchange, with the independent Taylor and analytic truncation checks on the preceding page.

The recorded environment is Python 3.14.5, NumPy 2.4.6, SymPy 1.14.0 and Matplotlib 3.11.0 on Windows. Parameter records, source hashes, predecessor hashes, two compilation passes and every inspected 95-dpi final render are bound by the stage evidence file. Full proofs are in the two companion notes on multi-source limits and actual-operator transfer. Earlier scientific mains are not rerun. This draft is saved locally, without a Git push or issue reply.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983, Secs. 3.2–3.3, 7 and 8.1.

[2] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584, Sec. 4.2, Eq. (4.10) and footnote 8.

[3] M. Junge and Q. Zeng, *Ultraproduct methods for mixed q-Gaussian algebras*, arXiv:1505.07852, Theorem 3.1 and Eq. (3.2).

[4] A. Day and Chengheng, local stages `susskind-singlet-multimark` and `susskind-mixed-cubic` (2026): charged common-disorder moments and unequal-probe kernel.

[5] A. Day and Chengheng, local stages `susskind-memory-two-component`, `susskind-memory-size-bridge` and `susskind-memory-joint-regression` (2026).

[6] A. Day and Chengheng, local stage `susskind-bilocal-reconstruction` (2026), with the force moments from `susskind-memory-return-expansion`.

[7] K. Itakura, *Boson Expansion Methods in (1+1)-dimensional Light-Front QCD*, arXiv:hep-th/9604032, Secs. 2, 4 and 5.